

# PRANOY RAY

Postdoctoral Researcher @ Argonne National Laboratory & The University of Chicago

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## PROFESSIONAL SUMMARY

Computational Chemist & Data Scientist with 5+ years of experience applying **advanced DOE strategies**, **Bayesian Optimization**, **Generative AI**, and **multiscale molecular simulations (DFT, AIMD, MD, DPD)** to accelerate molecular design and discovery. Proven track record in building **Inverse Design** and **Process Optimization** workflows that convert simulation and experimental data into manufacturable, high-yield chemical processes, cutting R&D trial-and-error iterations by up to 40% for Fortune 100 clients. Skilled at translating structure-property insight using/training **Surrogate Models (MLIPs)** and physics-informed ML into concrete engineering decisions, with direct experience leading cross-functional, multi-site R&D teams from lab-scale discovery through deployment.

## EDUCATION

- **Ph.D., Computational Materials Science** — Georgia Institute of Technology, Atlanta, USA (Aug 2021 – May 2026)
  - Advisor: [Dr. Surya R. Kalidindi](#) | Thesis: "On efficient screening strategies for molecular structures"
  - Minor: [Management of Technology \(MOT\)](#), Scheller College of Business (GT), Atlanta, USA
- **M.S., Computational Science and Engineering** — Georgia Institute of Technology, Atlanta, USA (2024)
- **B.Tech., Metallurgical & Materials Engineering** — National Institute of Technology, Durgapur, India (2020)

## TECHNICAL SKILLS

- **Multiscale Molecular Simulation:** DFT (VASP, PySCF, QuantumExpresso), Ab Initio MD (AIMD), Molecular Dynamics/DPD (LAMMPS, GROMACS, CG-Martini3), ML Interatomic Potentials (CHGNET/MACE/etc.), Force-Field Development, fine-tuning MLIPs.
- **ML & Generative AI:** PyTorch, Geometric Deep Learning, GNNs, CNNs, Autoencoders, Normalizing Flows, Scikit-learn, GPRs.
- **Process Optimization & Inverse Design:** Bayesian Optimization, Gaussian Processes (GPyTorch, BoTorch), Active Learning, Uncertainty Quantification (UQ), Design of Experiments (DoE), Pareto Frontier Analysis, FMEA.
- **Programming, HPC & Cloud:** Python (Advanced), R, C, Java, MPI, CUDA, AWS, GCP, Azure, SLURM/PBS, Flask, Shell Scripting.

## PROFESSIONAL EXPERIENCE

### Argonne National Laboratory & The University of Chicago

Postdoctoral Researcher

### Center for Organic Battery Innovation (Chicago, USA)

June 2026 – Present

- Developing data-efficient, active-learning-driven screening workflows that couple quantum-chemical calculations with surrogate ML models to prioritize redox-active organic candidates for next-generation battery materials.
- Building automated pipelines linking molecular simulation (DFT, MD) outputs to experimental validation, narrowing the candidate design space ahead of synthesis and cutting downstream screening effort+cost.
- Collaborating with national-lab, university, and cross-border teams (including Nobel laureates) to establish reproducible, high-throughput computational screening protocols for molecular discovery.

### Georgia Institute of Technology

Graduate Research Assistant (Advisor: [Dr. Surya R. Kalidindi](#))

### MINED Group (Atlanta, USA)

Aug 2021 – May 2026

- Engineered a Bayesian optimization framework to refine coarse-grained (CG) molecular topologies, performing inverse design for polymer miscibility in data-scarce regimes (published in *npj Computational Materials*).
- Built a voxel-based featurization engine paired with lean CNNs to map electron charge density fields to material properties, enabling rapid multi-element crystal screening and shortening DFT-workflow turnaround (published in *IMMI*; recognized as a **2025 TMS Standout Article**).
- Used Dissipative Particle Dynamics (DPD) and Molecular Dynamics to model microphase separation in block copolymers, integrating physical constraints with data-driven analysis to resolve mesoscale morphology.
- Led computational research on PFAS ("forever chemicals"), combining statistical mechanics and ML workflows to characterize helical structure and degradation mechanisms relevant to environmental remediation (published in *Advanced Science*).
- Served as Head TA for "ML Fundamentals for AI4Science" (ME8813 & ME4853), instructing and grading ~100 graduate and undergraduate students (Spring 2023 & Spring 2025).

### Multiscale Technologies Inc.

Data Science Manager & Materials Scientist Intern

### R&D Team (Seattle, USA)

Jan 2024 – Sep 2024

- Directed AI/ML strategy for semiconductor and specialty-chemical manufacturing clients (Fortune 100), architecting **Digital Twins** and physics-informed ML models that reduced experimental trial-and-error iterations by ~40%.
- Deployed Bayesian optimization and active-learning workflows to solve inverse design problems in manufacturing, translating thermodynamic and process data into engineering changes that improved yield.
- Architected the integration of physics-based simulations into the company's materials-design ecosystem, unifying disparate engineering APIs into automated analysis pipelines aligned with Root-Cause-Analysis and technology-management protocols.
- Recruited, managed, and mentored a distributed team of data scientists and engineers across the USA, France, India, and Pakistan, ensuring computational models met product-lifecycle and ROI requirements for client deployment.

### Bhabha Atomic Research Centre

Research Assistant (Advisors: [Dr. Srikumar Banerjee](#) & [Dr. Brahmananda Chakraborty](#))

### HP & SRPD (Mumbai, India)

May 2019 – Aug 2021

- Used DFT and Ab Initio Molecular Dynamics to theoretically discover and characterize three novel doped-carbon nanomaterial systems for hydrogen storage, translating first-principles results into design rules for alternative-fuel materials across three peer-reviewed publications.

- Led failure-analysis investigations on casting defects and implemented process-control measures that improved yield; applied Lean management principles to shop-floor quality control.

## SELECTED RESEARCH

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Author of 10+ peer-reviewed publications with 200+ citations — full record on [Google Scholar](#)  

### Selected Publications

- **ML Workflows for Screening Degradation-Relevant Properties of Forever Chemicals**  
P. Ray, A. Castillo, M. Kolel-Veetil, S.R. Kalidindi | **Advanced Science**, e23817 (Jan 2026) | [Code](#)
- **Refining Coarse-Grained Molecular Topologies: A Bayesian Optimization Approach**  
P. Ray, A.P. Generale, N. Vankireddy, Y. Asoma, M. Nakauchi, et. al. | **npj Computational Materials**, Vol. 11, Art. 234 (Jul 2025)
- **Electronic Manifolds for Extrapolative Alloy Discovery**  
P. Ray, S. Bhowmik, P. Suryanarayana, S.R. Kalidindi, A.J. Medford | **RSC Digital Discovery** (Accepted, July 2026) | [Code](#)
- **Spatial Statistics for Screening Molecular Structures**  
P. Ray, S.R. Kalidindi | arXiv preprint; **Small** (Under Review, May 2026)

### Selected Conference Talks

- **Statistical Quantification of Helicity in Linear PFAS** — ACS Spring Meeting, Atlanta, GA (Mar 2026)
- **(Invited) Lean CNNs for Materials Discovery Using Electron Charge Density Fields** — Trinity College, Dublin, Ireland (Host: Prof. Stefano Sanvito; Oct 2025)
- **Structure-Aware Bayesian Optimization for Efficient Design of Disordered CCAs** — SES Annual Technical Meeting, Atlanta, GA (Oct 2025)
- **(Invited) Bayesian Frameworks for Advanced Materials Design at the Atomistic Scale** — Novelis Global Research & Technology Center, Kennesaw, GA (Oct 2024)
- **(Invited) Bayesian Optimization of Coarse-Grained Topologies: Applications to Common Polymers** — TMS Fall Meeting, Pittsburgh, PA (Oct 2024)

## PROFESSIONAL SERVICE & LEADERSHIP

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- **Conference Peer Reviewer:** ICLR 2026, NeurIPS 2025, CVPR 2025, PEARC 2025, SciPy Conference (2023–2025); Session Chair, GT Undergraduate Research Symposium (2025).
- **Journal Service:** Editor, Integrating Materials and Manufacturing Innovation (Springer Nature); Reviewer, npj Computational Materials, Journal of Materials, Chemical Papers.
- **Leadership:** Board Member, Emerging Leaders Advisory Board @ Georgia Tech (2025–2026); Internal VP, Mechanical Engineering Graduate Association (MEGA) @ Georgia Tech (2023–2024).

## SELECTED AWARDS & ACCOLADES

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- **William H. Glenn Sr. Fellow** (2026): George W. Woodruff School of Mechanical Engineering, GeorgiaTech (Atlanta, GA, USA)
- **Woodruff School Fellow** (2025): Research Excellence, GWW School of Mechanical Engineering, GeorgiaTech (Atlanta, GA, USA)
- **Novelis Graduate Scholar** (2024): Novelis Innovation Hub & Novelis Inc. (Kennesaw, GA, USA)  
- Featured in: [GTRI News](#) as a top scholar advancing sustainability, circularity, and AI-driven materials discovery.
- **EIIF Fellow** (2022): TokyoTech & Strategic Energy Institute @GT (Honolulu, HI, USA)
- **CMS3 Fellow** (2022): NSF + Texas A&M University (College Station, TX, USA)
- **TMS Standout Article** (2025): Journal article on Lean CNNs highlighted by TMS Editors as a standout 2025 publication.
- **The Telegraph (India)** (2018): Featured in "Backyard Startups" for launching the bootstrapped venture #JustHashtags.